A Path-Minimum Conjecture for the Variance of Uniform Independent-Set Size in Trees

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Abstract

Choose an independent set uniformly from all independent sets of a tree, including the empty set. We conjecture that its size has uniquely minimum variance on the path of the same order. We give exact moment recursion, the path benchmark, a leaf bound, and a uniform proof for stars. For every fixed nonpath tree we prove strict path minimality at sufficiently small positive activity. The all-activity strengthening is false: for five vertices the star and path variances have exactly one positive crossover, between 12.799 and 12.800. We also prove the uniform inequality for an infinite branching family containing a 21-vertex obstruction to a natural grafting-descent lemma. Finite checks and archived extensions are distinguished from these proofs. The general activity-one conjecture remains open.

Preprint; not externally peer reviewed. AI use and verification are described at the end.

Introduction and research strategy

The formation notes identify a specific literature-led starting point: a known minimum theorem for the *average* size of a uniform independent set on trees. Instead of changing the graph family or sampling rule, the project changed the observable from mean to variance. This produces a natural adjacent problem, since a smaller mean need not imply smaller fluctuations. Equal order, uniform sampling, and inclusion of the empty set were fixed before comparing examples.

Small complete enumerations then tested the proposed minimizer, and searches of structured branching families attempted to disprove it. The much larger historical enumeration was accumulated later; it was not available at the initial formulation. The discovery route was therefore to move from an established extremal statistic to a different, logically independent one, then test its stability under branching and check whether existing results already implied it.

Research strategy. Partition-function derivatives retain the necessary first and second moments. Besides the uniform star theorem, we prove that every fixed nonpath tree has strictly larger variance than the path at sufficiently small positive activity. This local theorem does not extend to all activities: for five vertices the star and path variances have exactly one positive crossover, between 12.799 and 12.800. We also prove the uniform conjecture for an infinite branching family containing the 21-vertex local-grafting trap. Thus the main inequality can survive even where a proposed monotone simplification fails. These results isolate the pointwise activity-one comparison that still needs a general proof.

1 The probability model and conjecture

Let $\mathcal{I}(T)$ be the set of all independent vertex sets of a tree T, including $\varnothing$. Select I uniformly from $\mathcal{I}(T)$ and put $X_T = |I|$, $V(T) = \mathrm{Var}(X_T)$. The sampling is not uniform over maximal or

maximum independent sets, and is not a random greedy algorithm. Let P_n be the path on n vertices.

Conjecture 1.1. Every tree T of order $n \geq 1$ satisfies

$$V(T) \geq V(P_n),$$

with equality if and only if $T \cong P_n$.

The assertion is over unrooted trees; a root introduced for computation has no role in the probability distribution.

2 Existing research and what it does not settle

Andriantiana–Misanantenaina–Wagner prove that the path minimizes the *mean* independent-set size among trees of a given order [1, Theorem 4.4]. A mean inequality does not imply a variance inequality. Davies–Sandhu–Tan study expectations and variances in the hard-core model [2]; Zhang–Xu subsequently give degree-dependent variance-fraction lower bounds [3]. For example, their stated bound at activity one gives total variance at least $n/(\Delta+2)^2$ for maximum degree Δ. This is too weak to recover the path's asymptotic coefficient $1/(5\sqrt{5})$, even when $\Delta = 2$. Levit's recent forest coefficient work also uses conditional variance estimates [4]; coefficient-shape statements are distinct from the exact fixed-order minimum in Conjecture 1.1.

The hard-core model assigns probability proportional to $\lambda^{|I|}$; uniform sampling is exactly $\lambda = 1$. The definitions, moment derivatives and conditional-independence methods are established. Focused searches through 14 November 2026 did not locate the exact conjecture in the checked sources. This limited search is not a priority guarantee, and the auxiliary estimates below are not claimed to be historically new merely because they are derived here.

3 Exact moments and the path benchmark

Write

$$Z_T(z) = \sum_{I \in \mathcal{I}(T)} z^{|I|}, \qquad D = z\frac{d}{dz}.$$

At activity z, $\mathbb{E}_z X_T = D \log Z_T(z)$ and $\operatorname{Var}_z X_T = D^2 \log Z_T(z)$, by differentiating the finite sum. At $z = 1$, use the integer triple (Z, M, Q) with $M = \sum_I |I|$ and $Q = \sum_I |I|^2$. Independent products combine triples by

$$(Z_1 Z_2,\ M_1 Z_2 + Z_1 M_2,\ Q_1 Z_2 + 2M_1 M_2 + Z_1 Q_2).$$

At a rooted vertex, its excluded state multiplies the total child states, while its included state multiplies excluded child states and then adds one to the size. Adding these two triples gives the full triple. Thus

$$V(T) = \frac{QZ - M^2}{Z^2}$$

can be compared using integers, with no floating-point decision.

For a path let $f_n(z) = Z_{P_n}(z)$. Then $f_0 = 1$, $f_1 = 1 + z$, and

$$f_n = f_{n-1} + z f_{n-2} = \frac{\rho_+^{n+2} - \rho_-^{n+2}}{\sqrt{1+4z}}, \qquad \rho_\pm = \frac{1 \pm \sqrt{1+4z}}{2}.$$

Proposition 3.1. With $c = 1/(5\sqrt{5})$ and $\beta = 2c - 2/25$,

$$V(P_n) = cn + \beta + \varepsilon_n, \qquad |\varepsilon_n| \le R(n+2),$$

where

$$R(t) = \left(\frac{2}{5}\right)^t \left(\frac{t}{5(1-(2/5)^t)} + \frac{t^2}{5(1-(2/5)^t)^2}\right).$$

In particular $V(P_n) < 0.09n + 0.11$ for $n \ge 6$.

Proof. Put $t = n+2$ and $q = \rho_-/\rho_+$. Then $\log f_n = t\log\rho_+ - \frac{1}{2}\log(1+4z) + \log(1-q^t)$. At $z = 1$, $D^2 \log\rho_+ = c$ and $D^2[\frac{1}{2}\log(1+4z)] = 2/25$. Also $|q| = (3-\sqrt{5})/2 < 2/5$, $D\log|q| = 1/\sqrt{5}$ and $D^2\log|q| = -2/(5\sqrt{5})$. Twice differentiating the last logarithm and taking absolute values gives the displayed bound, using $2/(5\sqrt{5}) < 1/5$. The factors $t(2/5)^t$ and $t^2(2/5)^t$ decrease for integer $t \ge 2$, as do the reciprocal denominator factors. Hence $R(t) \le R(8) < 0.01$ for $t \ge 8$. Now $c < 0.09$ and $\beta < 0.10$ give the final claim. □

4 A leaf bound and an infinite family

Theorem 4.1. For a tree of order at least three, let k_v be the number of leaf neighbors of a nonleaf vertex v. Then

$$V(T) \ge \sum_{v:k_v>0} \frac{k_v 2^{k_v}}{4(1+2^{k_v})}.$$

Proof. Condition on all nonleaf occupancies. Leaves at an unoccupied parent are independent fair choices, and leaves at an occupied parent are forced absent. Conditional variance is therefore one quarter of the number of leaves with unoccupied parents. The law of total variance gives $V(T) \ge \frac{1}{4}\sum_v k_v \Pr(v \notin I)$. For each independent set containing v, remove v and choose an arbitrary subset of its k_v leaves. This gives 2^{k_v} distinct sets avoiding v; the map is injective, since all outside choices are retained. Thus $\Pr(v \notin I) \ge 2^{k_v}/(1+2^{k_v})$, proving the bound. □

Theorem 4.2. For every star $K_{1,n-1}$ with $n \ge 4$, $V(K_{1,n-1}) > V(P_n)$.

Proof. For $n \ge 6$, Theorem 4.1 gives

$$V(K_{1,n-1}) \ge \frac{(n-1)2^{n-1}}{4(1+2^{n-1})} \ge \frac{8(n-1)}{33} > 0.09n + 0.11 > V(P_n),$$

where the strict middle difference is $(503n - 1163)/3300 > 0$. For $n = 4$, the star variance is $56/81 > 7/16$; for $n = 5$, it is $288/289 > 94/169$. Orders at most three are already paths. □

The leaf bound is useful for leaf-rich trees but loses the correlations needed when leaves are far from the branching core. It is not a sharp universal path comparison.

5 Two false strengthenings and their lessons

5.1 The activity cannot be arbitrary

The assertion $\mathrm{Var}_\lambda(X_T) \ge \mathrm{Var}_\lambda(X_{P_n})$ for every $\lambda > 0$ is false.

Proposition 5.1. For every odd $n = 2m+1 \ge 5$, the star has smaller variance than the path at all sufficiently large activities.

Proof. If a polynomial has top terms $az^\alpha + bz^{\alpha-1}$, then $D^2 \log Z(z) = (b/a)z^{-1} + O(z^{-2})$ as $z \to \infty$. The number of size-k independent sets of a path is $\binom{n-k+1}{k}$. Thus for P_{2m+1} the leading two coefficients are 1 and $\binom{m+2}{2}$. The star polynomial is $(1+z)^{2m} + z$, whose leading two coefficients are 1 and $2m$ for $m \geq 2$. Since

$$\binom{m+2}{2} - 2m = \frac{m^2 - m + 2}{2} > 0,$$

the star has the smaller leading variance term, proving the claim. □

For instance, $Z_{P_5} = 1 + 5z + 6z^2 + z^3$ and $Z_{K_{1,4}} = 1 + 5z + 6z^2 + 4z^3 + z^4$. At $z = 1$ their variances are $94/169 < 288/289$, while at $z = 100$ they are approximately $0.05516 > 0.03922$. This refutes the activity strengthening, not Conjecture 1.1.

5.2 A single-arm descent can be trapped

Consider the 21-vertex tree T_* consisting of a center joined to four vertices, each of which supports two pendant paths of length two. A proposed simplification move detaches one of these eight length-two arms from its branch vertex and reattaches it at any leaf outside that arm. There are $8 \cdot 7 = 56$ such moves. Every one *increases* variance.

Here is an exact verification independent of any picture. Using the path polynomials above, the original independence polynomial is

$$Z_* = f_5^4 + z f_2^8.$$

The moved trees have two symmetry types: eight reattach at the other leaf of the same branch, with polynomial $f_5^4 + z f_4 f_2^6$; the other 48 have polynomial $f_3 f_7 f_5^2 + z f_4 f_2^6$. Their triples (Z, M, Q) are respectively

$$(35122, 217313, 1419977), \quad (34393, 212210, 1383284), \quad (34562, 213575, 1394497).$$

Substitution in $(QZ - M^2)/Z^2$ gives the positive variance increases

$$\frac{4284827947109583}{1459145487942294916}, \qquad \frac{1432841417801136}{92095036891232881}.$$

Yet $V(T_*) - V(P_{21}) = 171248959884482305/1013024443080251716 > 0$. Thus T_* is an obstruction to this particular local-descent proof, not a counterexample to the main conjecture. The supplied program also constructs all 56 moves directly and checks these two exact increases.

6 Local comparison theorems beyond activity one

6.1 Every fixed nonpath tree loses near activity zero

Let d_v be the degrees of T, and define its branching surplus by

$$b(T) = \sum_{v : d_v \geq 3} \binom{d_v - 1}{2}.$$

It is positive precisely when the tree is not a path.

Theorem 6.1. For every fixed nonpath tree T of order n,

$$\mathrm{Var}_\lambda(X_T) - \mathrm{Var}_\lambda(X_{P_n}) = 9b(T)\lambda^3 + O_T(\lambda^4) \qquad (\lambda \downarrow 0).$$

In particular, there exists $\varepsilon_T > 0$ such that the path has strictly smaller variance for every $0 < \lambda < \varepsilon_T$.

Proof. The first two nonconstant coefficients of a tree's independence polynomial are fixed by its order:

$$i_1(T) = n, \qquad i_2(T) = \binom{n}{2} - (n-1).$$

To count independent triples, start with all triples and exclude those containing an edge. Each of the $n-1$ edges lies in $n-2$ triples. A pair of edges can lie in the same triple only when they share a vertex, and a tree has no triangle. Inclusion-exclusion therefore gives

$$i_3(T) = \binom{n}{3} - (n-1)(n-2) + \sum_v \binom{d_v}{2}.$$

For the path the last sum is $n-2$, while the degree sum identity gives

$$\sum_v \binom{d_v}{2} - (n-2) = \frac{1}{2}\sum_v (d_v - 1)(d_v - 2) = b(T).$$

The first two coefficients of Z_T and Z_{P_n} agree, so

$$\log Z_T(\lambda) - \log Z_{P_n}(\lambda) = b(T)\lambda^3 + O_T(\lambda^4).$$

Applying $D^2 = (\lambda\, d/d\lambda)^2$ multiplies the cubic coefficient by nine. Positivity of $b(T)$ proves the final assertion. □

This is a local theorem for every fixed tree, with no degree restriction. It does not provide an ε_T reaching one, or a bound uniform over all orders. Together with Proposition 5.1, it shows that some tree comparisons must change sign as activity increases.

The known mean theorem also cannot bridge that gap by differentiation. In fact, writing $\Delta V(\lambda) = \mathrm{Var}_\lambda(X_T) - \mathrm{Var}_\lambda(X_{P_n})$,

$$\mathbb{E}_1 X_T - \mathbb{E}_1 X_{P_n} = \int_0^1 \frac{\Delta V(\lambda)}{\lambda}\, d\lambda.$$

This follows from $\lambda(d/d\lambda)\mathbb{E}_\lambda X = \mathrm{Var}_\lambda X$ and both means tending to zero at zero. A nonnegative integral does not determine the value of its integrand at one.

6.2 A complete five-vertex star–path comparison

For the five-vertex polynomials given above, direct differentiation and multiplication give

$$\mathrm{Var}_\lambda(X_{K_{1,4}}) - \mathrm{Var}_\lambda(X_{P_5}) = \frac{\lambda^3 Q(\lambda)}{Z_{K_{1,4}}(\lambda)^2 Z_{P_5}(\lambda)^2},$$

where

$$\begin{aligned} Q(x) = {} & 27 + 316x + 1518x^2 + 3908x^3 + 5922x^4 \\ & + 5496x^5 + 3111x^6 + 1032x^7 + 174x^8 + 4x^9 - 2x^{10}. \end{aligned}$$

Proposition 6.2. There is exactly one positive activity λ_* at which the two variances agree, and

$$12.799 < \lambda_* < 12.800.$$

The star has larger variance below λ_* and smaller variance above it.

Proof. The function $Q(x)/x^{10}$ is a sum of strictly positive multiples of negative powers of x, minus two. It is strictly decreasing on $(0,\infty)$, tends to $+\infty$ at zero, and tends to -2 at infinity. Thus it has exactly one zero. The denominator in the variance formula is positive. Exact rational evaluation gives $Q(12799/1000) > 0$ and $Q(64/5) < 0$, proving the stated interval and signs. □

The cubic term $27\lambda^3$ agrees with Theorem 6.1, since the star has $b = 3$. The negative highest coefficient agrees with the large-activity difference $-2/\lambda + O(\lambda^{-2})$. These are independent checks of the local and tail mechanisms, not a numerical fit for the crossing.

6.3 An infinite family containing the grafting trap

Let T_m have a center adjacent to m vertices, each of which supports two pendant arms of length two. It has $5m + 1$ vertices and

$$Z_{T_m}(z) = f_5(z)^m + z f_2(z)^{2m}.$$

In particular $T_4 = T_*$ from the failed local-descent rule.

Theorem 6.3. For every $m \geq 1$, $V(T_m) > V(P_{5m+1})$.

Proof. At activity one, the probability that the center is selected is

$$p_m = \frac{9^m}{13^m + 9^m}.$$

Given that it is absent, the independent components are m copies of P_5, with mean $20m/13$ and variance $94m/169$. Given that it is present, there are $2m$ copies of P_2 and the selected center, with mean $1 + 4m/3$ and variance $4m/9$. Total variance gives the exact formula

$$V(T_m) = (1 - p_m)\frac{94m}{169} + p_m\frac{4m}{9} + p_m(1 - p_m)\left(\frac{8m}{39} - 1\right)^2.$$

For $m \geq 3$, $(9/13)^m \leq 729/2197 < 1/3$, so $p_m < 1/4$. Since $94/169 > 4/9$,

$$V(T_m) > \frac{1607m}{3042} > \frac{9m}{20} + \frac{1}{5} > V(P_{5m+1}).$$

The last inequality is Proposition 3.1; the middle difference is $(2381m - 6084)/30420 > 0$ for $m \geq 3$. For the two remaining cases direct substitution gives

$$V(T_1) = \frac{321}{484}, \quad V(T_1) - V(P_6) = \frac{7009}{213444} > 0,$$

and

$$V(T_2) = \frac{69761}{62500}, \quad V(T_2) - V(P_{11}) = \frac{113004929}{3393062500} > 0.$$

□

Thus a whole explicitly controlled family satisfies the conjecture even though one member blocks every allowed single-arm descent. A global proof need not, and for that move cannot, be assembled from a strict decrease at each local simplification.

The supplied extension checker verifies the independent-triple coefficient on all 201 trees through order 10, constructs T_m for $1 \leq m \leq 30$ independently of its polynomial formula, and checks the crossover numerator and its rational sign bracket. These checks accompany the displayed proofs; their finite limits are not premises for the infinite theorems.

7 Finite evidence, archived advances, and the remaining task

Fresh enumeration generates every unlabelled tree through order 10 by adding a leaf and deduplicating canonical encodings. Every tree can be obtained by deleting a leaf from a smaller tree, so induction ensures coverage. The respective numbers are

$$1, 1, 1, 2, 3, 6, 11, 23, 47, 106,$$

totalling 201 trees. For each tree, integer-moment recursion agrees with direct enumeration of all vertex subsets. Every order has unique equality at the path. The script `verify_trees.py` and `tree_verification.json` also check the leaf bound, stars through order 200, the activity counterexample, and the grafting trap.

The older local research archive reports a completed computer-assisted verification through order 30, covering 23522619475 unlabelled trees, with some order-30 cases excluded by certified strict bounds. Its orders 31–100 result covers only trees with at most five leaves, namely 2114296960 nonpaths plus 70 path benchmarks. It must not be described as enumeration of *all* trees through order 100. The archive also contains separate arguments for spiders, trees with at most six leaves, trees with at most three branch vertices, and several classes of branching cores. These are archived extensions, not self-contained theorems reproved in this note; the small fresh checks above do not validate billions of earlier cases. The accompanying research audit points to the original proof notes and completed-run records.

The unresolved task is a sharp comparison for arbitrary branching arrangements, especially several interacting branch vertices joined by short paths. A proof cannot simply differentiate the known mean inequality, extend the statement to every activity, or insist that each single-arm move decreases variance. Promising approaches must retain several boundary moments at once, or produce a global decomposition with controlled error and a strict equality analysis. Existing infinite families and large finite evidence leave that structural problem intact.

Provenance and use of AI

This is a research draft, not a peer-reviewed publication. The problem arose in an AI-assisted exploratory project. AI tools were used substantially for literature searches, mathematical derivations, verification programs, and drafting. The proofs and bounded checks are supplied for independent scrutiny; neither an exhaustive novelty claim nor a general proof beyond the stated results is asserted.

References

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